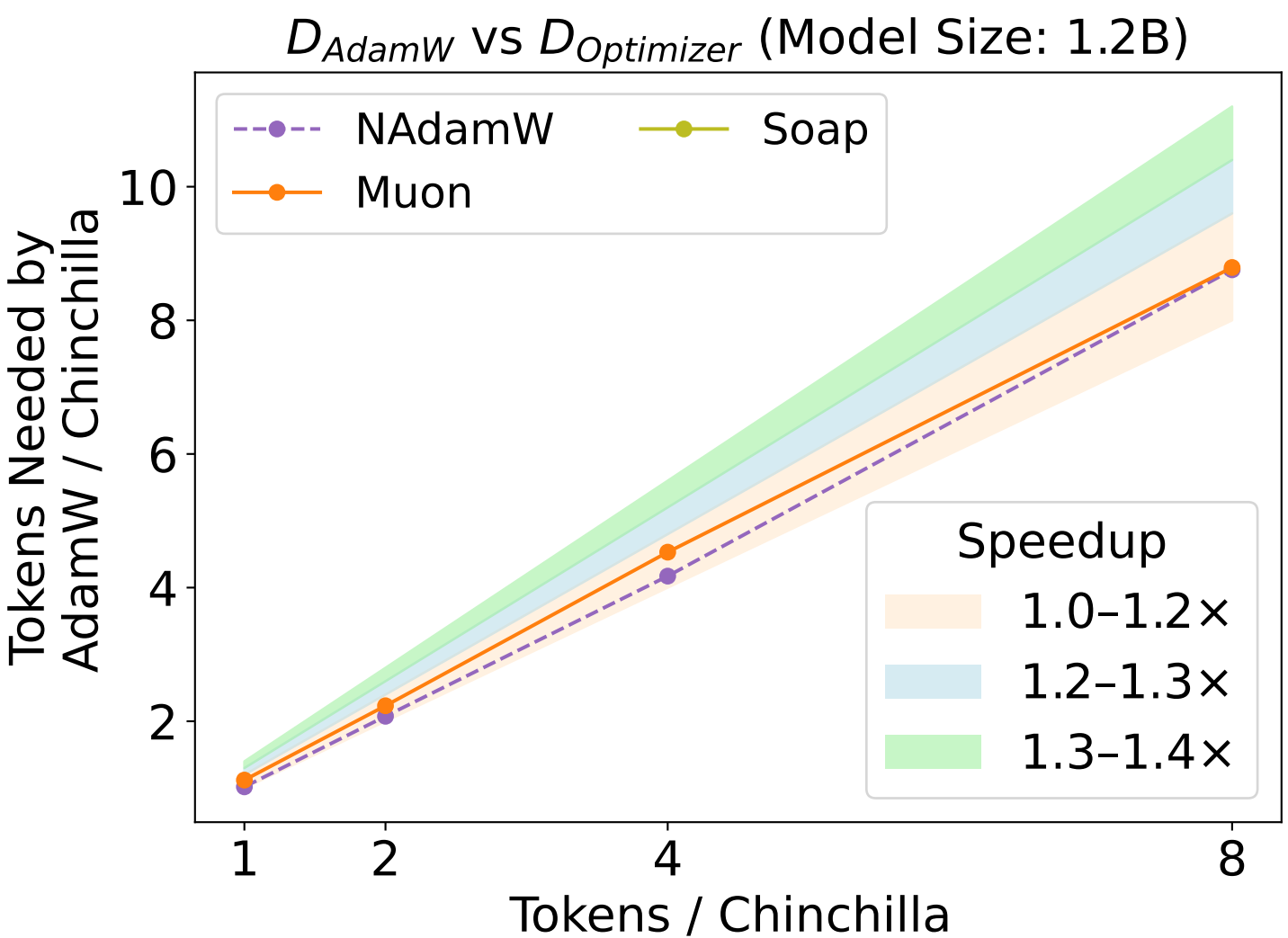
$D_{AdamW}$ vs $D_{Optimizer}$ (Model Size: 1.2B)
NAdamW
Soap
Muon
Speedup
1.0–1.2×
1.2–1.3×
1.3–1.4×
Tokens Needed by AdamW / Chinchilla
Tokens / Chinchilla
2
4
6
8
10
1
2
4
8